

Notes: Faber (RES, 2014)

Trade Integration, Market Size, and Industrialization: Evidence from China's National Trunk Highway System

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1 Big picture

- **Question.** When a large transport network lowers trade costs between very large metropolitan markets and much smaller peripheral regions, does economic activity diffuse to the periphery or become even more concentrated in the core?
- **Setting.** The paper studies China's National Trunk Highway System (NTHS), a huge expressway program approved in 1992 and largely built between 1998 and 2003. The network was designed to connect provincial capitals, large cities, and border crossings on a single expressway grid.
- **Why this setting is powerful.** The targeted metropolitan centers represented only about 1.5% of China's land area and 17% of its population, but they accounted for roughly 60% of nonagricultural production before the bulk of the network was built. The average preexisting market-size gap between targeted centers and non-targeted peripheral counties was about 25 to 1 in county GDP.
- **Core empirical challenge.** Route placement was not random. Planners favored politically important and economically prosperous places along the way between targeted nodes. So a simple comparison of connected and non-connected counties would be biased.
- **Main conceptual contribution.** The paper brings together transport-infrastructure evaluation and trade/economic-geography theory. It asks not just whether roads raise activity, but whether lowering trade costs between highly asymmetric markets can actually hurt smaller peripheral producers.
- **Main empirical design.** Faber compares peripheral counties connected to the NTHS with peripheral counties not connected to it, instrumenting actual exposure with hypothetical minimum spanning tree networks: a least-cost-path spanning tree and a Euclidean spanning tree.
- **Main baseline result.** NTHS connections reduced GDP growth, industrial output growth, nonagricultural output growth, and local government revenue growth in connected peripheral counties relative to non-connected peripheral counties.
- **Economic magnitudes.** In the preferred specification with both instruments and the full control set, connected peripheral counties experienced roughly an 18% lower GDP growth rate, a 23% lower local-government-revenue growth rate, and a 26% lower industrial-output growth rate over 1997–2006.
- **Key channel result.** The evidence favors a *trade-based core-periphery mechanism* rather than a local decentralization or suburbanization mechanism. Connected peripheral counties do not become more urbanized, and neighboring counties do not gain enough to support a decentralization story.
- **Key heterogeneity result.** The adverse effect is weaker for peripheral counties that were initially larger and for counties that initially faced higher trade costs to the targeted metropolitan core. That is exactly what a core-periphery trade model would predict.
- **Main credibility result.** The paper combines a strong first stage, rich county controls, placebo tests using the pre-network period, and many appendix robustness checks to argue that the estimates capture the effects of NTHS-induced trade integration rather than omitted local trends.

- **Economic takeaway.** Large inter-regional transport projects can create national efficiency gains while still worsening relative outcomes in connected peripheral regions. Better infrastructure does not automatically imply diffusion of industrial activity to poorer places.

2 Data and measurement (key objects)

2.1 Why China and the NTHS are a useful laboratory

- The NTHS created a sharp change in within-country trade integration at a national scale.
- The policy objective was explicitly to connect *targeted* metropolitan nodes, not to help peripheral counties directly.
- That feature generates a clean question: what happened to the smaller counties that happened to lie on the routes between big cities?
- Because market-size differences were large ex ante, the setting is especially well suited to testing core-periphery predictions from trade theory.

Interpretation. The empirical design exploits a policy that was national in purpose but locally uneven in its exposure. That is exactly what is needed to study whether the periphery gains or loses when connected to the core.

2.2 Policy background and timing

- In 1992, the State Council approved the “7–5” network: seven horizontal and five vertical axes.
- The network ultimately spanned about 35,000 km of four-lane limited-access highways at a cost of roughly US\$120 billion.
- Construction is split into a small “kick-off” phase from 1992 to 1997 and a “rapid development” phase from 1998 to 2007.
- About 10% of NTHS routes were open before mid-1997, 81% opened between mid-1997 and the end of 2003, and the remaining 9% opened after 2003.
- The paper’s main variation comes from connections completed by the end of 2003 and their effects on county outcomes between 1997 and 2006.

Interpretation. The fact that the bulk of the network was built after the 1997 baseline year makes the design close to a before-after comparison around a major infrastructure shock.

2.3 Data sources and sample construction

The paper combines several county-level data sources:

- 1999 geo-referenced administrative boundaries from the ACASIAN Data Center;
- county socioeconomic records from Provincial Statistical Yearbooks for 1990, 1997, and 2006;

- the 1990 Chinese population census;
- geo-referenced NTHS route maps digitized from high-resolution road atlases;
- land cover, hydrology, and elevation data used to build least-cost paths.

The geographical base contains 2,341 county-level units, 349 prefectures, and 33 provinces. The final usable county database contains 1,748 historically consistent geo-referenced county units with non-missing 1997 and 2006 socioeconomic data. Of these, 1,706 also report 1990 census information, and 1,238 report local government revenues in 1990.

Interpretation. The paper works with a very large county panel for a developing-country setting, and it observes multiple outcomes, multiple baseline years, and detailed pre-network county characteristics.

2.4 Treatment variables: what it means to be connected

The main binary treatment is

$$Connect_{ip} = 1$$

if any part of county i in province p lies within 10 km of an NTHS segment that was open to traffic by the end of 2003.

The paper also uses a continuous treatment:

$$\ln DistHwy_{ip},$$

which is the logarithm of the great-circle distance from the county center to the nearest NTHS segment opened before the end of 2003.

Two sample restrictions are important:

- the analysis focuses on **peripheral counties**;
- counties within a 50 km commuting buffer around targeted city centers are excluded so that the paper does not mix within-metropolitan effects with inter-regional trade effects.

Interpretation. The treatment is not “being near a city.” It is being newly connected to a national expressway network while remaining outside the commuting zone of the targeted metropolitan center.

2.5 Outcome variables

The paper studies changes between 1997 and 2006 in the logarithms of:

- industrial gross value added, $\ln(IndGVA)$;
- nonagricultural gross value added, $\ln(NonAgGVA)$;
- total GDP, $\ln(GDP)$;
- local government revenue, $\ln(GovRevenue)$;
- agricultural gross value added, $\ln(AgGVA)$;

- registered population, $\ln(Population)$.

For mechanism tests, the paper also studies:

- urban population growth;
- changes in urbanization;
- and spatial gradients in the effect with respect to distance from the highway.

Interpretation. The outcome set is built to answer three different questions at once: total economic activity, sectoral composition, and the demographic/urbanization channel.

2.6 County controls

The main control vector contains preexisting county characteristics designed to block the most obvious violations of the exclusion restriction:

- $\ln DistNode$: log distance to the nearest targeted metropolitan node;
- a dummy for prefecture-capital status in 1990;
- a dummy for city status in 1990;
- $\ln UrbPop90$: log urban registered population in 1990;
- $Educ90$: the share of above-compulsory schooling in the population aged 20 and older in 1990;
- $AgShare90$: the share of agricultural employment in 1990.

Interpretation. These variables aim to absorb exactly the kinds of preexisting political and economic differences that might have led planners to place routes through some peripheral counties rather than others.

2.7 Descriptive facts

Table 1 shows strong ex ante differences across county groups in 1997:

- targeted city centers are vastly larger and richer than peripheral counties;
- connected peripheral counties are richer, more urbanized, and more industrialized than non-connected peripheral counties even before the network is built;
- mean GDP is 517.86 hundred million yuan in targeted city centers, 32.58 in connected peripheral counties, and 15.09 in non-connected peripheral counties;
- the number of counties is 54 targeted city-center units, 424 connected peripheral counties, and 943 non-connected peripheral counties.

Interpretation. The raw data already show that connected counties were positively selected. That is why the IV strategy is not just a refinement but the core of the paper.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and competing hypotheses

The paper contrasts two broad stories.

Trade-based core-periphery channel. If trade costs between big metropolitan markets and small peripheral counties fall, industrial production may become more concentrated in the large markets. This can be rationalized by:

- increasing returns and the home-market effect;
- economic-geography agglomeration forces;
- or comparative-advantage/factor-mobility stories with frictions.

Local decentralization channel. An alternative story is that the highway induces local decentralization: activity spreads from connected peripheral counties to their nearby neighbors, or the instrument accidentally picks up second-tier city dynamics rather than inter-regional trade effects.

Testable implications.

- A trade-based channel predicts weaker adverse effects for larger peripheral counties and for counties that initially faced higher trade costs to the core.
- A decentralization channel predicts more urbanization in connected counties and positive spillovers to neighboring peripheral counties.

Interpretation. The paper is not simply measuring a reduced-form road effect. It is using the sign pattern and heterogeneity of that reduced form to distinguish between very different mechanisms.

3.2 Baseline difference-in-differences specification

The core specification is

$$\ln y_{ip,2006} - \ln y_{ip,1997} = \gamma_p + \beta \text{Connect}_{ip} + \eta X_{ip} + \varepsilon_{ip},$$

where:

- y_{ip} is a county-level outcome;
- γ_p is a province fixed effect;
- Connect_{ip} is the NTHS connection indicator;
- X_{ip} is the vector of pre-1990 county controls.

The alternative continuous-treatment version replaces the connection dummy with $\ln \text{DistHwy}_{ip}$.

Interpretation. This specification compares growth between 1997 and 2006 across otherwise similar peripheral counties, asking whether connected counties grow differently once broad province effects and baseline county traits are netted out.

3.3 Least-cost-path and Euclidean spanning tree instruments

The paper instruments for actual route placement with two hypothetical networks.

1. Least-cost-path spanning tree.

- Step 1: use land cover, elevation, and hydrology data plus Dijkstra’s algorithm to compute least-cost bilateral routes between targeted nodes.
- Step 2: use those bilateral costs inside Kruskal’s minimum spanning tree algorithm to choose the set of links that connects all targeted nodes at minimum total construction cost.

2. Euclidean spanning tree.

- Construct great-circle distances between targeted nodes.
- Run Kruskal’s algorithm on those distances.
- Add extra routes by repeating the procedure in North-Centre-South and East-Centre-West subdivisions to better match the fact that the actual NTHS built more than the minimum number of spanning-tree links.

Interpretation. The first instrument is more geographically realistic at the route level. The second is cruder but sometimes captures more of the actual network. Using both together is meant to exploit their complementary strengths.

3.4 Estimation details and standard errors

- Baseline regressions are estimated by OLS for reduced form and 2SLS for IV.
- All regressions include province fixed effects.
- Standard errors are clustered at the province level, reflecting the fact that network shocks may be correlated within provinces.
- The paper also reports Conley-style standard errors that allow spatial dependence to decline with geographic distance.

Interpretation. The identifying variation is cross-county within China, but the paper is careful not to overstate precision when nearby counties may be jointly affected by related route placements.

3.5 First stage

Table 2 shows that both instruments strongly predict NTHS exposure.

- For the binary connection measure, the least-cost-path instrument enters around 0.25–0.32 and the Euclidean instrument around 0.14–0.24.
- The first-stage F-statistics range from about 20 to 32 in the main connection regressions and remain above 15 even in the distance specifications.
- Counties closer to targeted nodes, with city status, and with larger 1990 urban populations were more likely to be connected.

Interpretation. The instruments are relevant, and the first stage also reinforces the paper’s point that actual planners targeted more prosperous places—which is exactly why IV matters.

3.6 Baseline reduced-form and IV results

Table 3 is the heart of the paper. The preferred specification uses both instruments and the full control set.

Main coefficients:

- $\Delta \ln(\text{IndGVA})$: about -0.297 ;
- $\Delta \ln(\text{NonAgGVA})$: about -0.268 ;
- $\Delta \ln(\text{GovRevenue})$: about -0.257 ;
- $\Delta \ln(\text{GDP})$: about -0.203 ;
- $\Delta \ln(\text{AgGVA})$: close to zero and insignificant;
- $\Delta \ln(\text{Population})$: close to zero and insignificant.

Two important patterns stand out.

- IV estimates are more negative than OLS estimates, consistent with positive selection into actual road placement.
- Adding county controls pushes the IV estimates further negative, which strengthens the interpretation that connected counties were ex ante better positioned to grow.

Interpretation. The roads do not simply fail to help the periphery; conditional on the paper’s identifying assumptions, they actually lower relative industrial and GDP growth in connected peripheral counties.

3.7 Placebo and falsification specification

The paper exploits the fact that many counties have data before the network was in place and estimates the same type of specification on local-government-revenue growth over 1990–1997 and 1997–2006.

The key pattern from Table 4 is:

- no significant effect before the network is built;
- a negative and significant effect only in the actual NTHS period.

For example, in the specification using both IVs and the binary connection measure, the coefficient is about 0.056 for 1990–1997 and about -0.204 for 1997–2006.

Interpretation. This is a direct test of the exclusion restriction. If spanning-tree locations were simply counties with bad underlying trends, the paper should have found a pre-NTHS effect too. It does not.

3.8 Mechanism tests: urbanization, neighbors, and distance gradients

To distinguish trade integration from local decentralization, the paper estimates several additional specifications:

- outcomes on urban population and urbanization;
- outcomes with a *neighbor* dummy for peripheral counties adjacent to a connected county;
- outcomes on continuous distance to the nearest NTHS segment.

Main findings from Table 5 and Figure 4:

- connected counties do not experience significant urban-population or urbanization gains;
- adding the neighbor dummy barely changes the negative coefficient on connected counties;
- greater distance to the NTHS is associated with better industrial, GDP, and government-revenue growth.

Interpretation. The negative effects are concentrated in connected counties themselves and fade with distance. That pattern fits a trade-integration story much better than a local suburbanization story.

3.9 Heterogeneity specification

To test the core-periphery mechanism more directly, the paper augments the baseline with interactions:

$$\Delta \ln y_{ip} = \alpha + \beta \text{Connect}_{ip} + \delta (\text{Connect}_{ip} \times \ln \text{DistNode}_{ip}) + \phi (\text{Connect}_{ip} \times \text{Emp90Dum}_{ip}) + \eta X_{ip} + \varepsilon_{ip}.$$

There is also a continuous-treatment analogue using $\ln \text{DistHwy}$ and its interactions.

Table 6 shows:

- positive and significant interaction terms for $\text{Connect} \times \ln \text{DistNode}$ and $\text{Connect} \times \text{Emp90Dum}$;
- negative and significant interactions for the continuous-distance version.

Interpretation. The adverse connection effect is weaker for counties that were initially larger or more remote from the core. That is exactly the comparative-static prediction of a home-market/core-periphery model.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline estimate

Identification rests on the claim that, conditional on province fixed effects and rich preexisting county controls, location on a hypothetical minimum spanning tree affects later county growth only

through actual NTHS exposure.

The empirical design is credible because:

- the network's targeted nodes were chosen nationally;
- peripheral counties were not the stated beneficiaries of the program;
- and the proposed instruments are built from engineering-style cost minimization rather than from economic outcomes.

4.2 Why route placement is not random and why IV is necessary

The paper is explicit that OLS would be hard to interpret.

- Connected peripheral counties were richer, larger, more urbanized, and more industrialized even before the network was built.
- Province and county governments financed much of the construction by borrowing against future toll revenues, which created incentives to favor places expected to attract traffic.

Interpretation. Because prosperous counties were more likely to be placed on actual routes, any simple connected-versus-not comparison would likely understate how adverse the true causal effect was.

4.3 Why reverse causality is weak

The main source of variation is whether a county lies near a highway built between 1992 and 2003, while the outcomes are measured as growth from 1997 to 2006. Since future county growth could not cause the 1992 network blueprint, and the placebo period shows no pre-network effect, reverse causality is not the natural concern here. The real issue is omitted route-selection bias, which the IV and placebo strategy address.

4.4 How fixed effects and controls absorb common unobservables

- Province fixed effects absorb broad regional differences in growth, institutions, and development.
- In *DistNode* absorbs the mechanical tendency for counties nearer to large targeted cities to lie on spanning-tree routes.
- Political and economic controls absorb the possibility that counties on natural corridors were historically more urban, more skilled, or more administratively important.

Interpretation. The identification does not ask whether connected and non-connected counties are identical in levels. It asks whether, after conditioning on these observables, residual variation in cost-minimizing route exposure is plausibly exogenous.

4.5 Robustness checks from the paper and appendix

The paper reports that the main effects are robust to controlling for or excluding:

- the geography used to build the least-cost-path instrument;
- construction already under way by 1997;
- mountainous regions;
- regions affected by the Golmud–Lasa railway;
- proximity to historical trade routes such as the Silk Road;
- 1997 market-access differences;
- county proximity to the coastline;
- and sample-selection concerns from boundary changes.

Interpretation. This matters because the paper’s story is subtle: the roads lower trade costs, but the reduced form is negative in the periphery. The robustness battery is meant to show this is not a fragile artifact of one particular map or control choice.

4.6 Strengths and limitations of the instrument set

Strengths.

- The least-cost-path IV is grounded in explicit route-construction costs.
- The Euclidean IV broadens coverage of actual bilateral links.
- The first stage is comfortably strong in the main specifications.

Limitations.

- In the interaction specifications, the first-stage F-statistics fall sharply.
- The paper cannot fully distinguish among all trade microfoundations: increasing returns, comparative advantage with frictions, and capital reallocation can all rationalize a negative peripheral effect.
- The setting is not designed to estimate the aggregate national welfare effect of the NTHS.

Interpretation. The paper is strongest on the sign, magnitude, and pattern of the peripheral reduced form. It is more modest about the exact structural mechanism and about national welfare.

4.7 Relation to the literature

The paper contributes to several literatures at once:

- transport-infrastructure evaluation;
- economic geography and home-market effects;
- and trade/infrastructure work in China.

Relative to Banerjee, Duflo, and Qian, and to Baum-Snow and coauthors, Faber’s distinctive contribution is to focus directly on asymmetric market size and to build an IV from minimum spanning tree networks rather than from simple straight-line historical connections.

5 Tables and figures

5.1 Figure 1 and Table 1: the network and the initial asymmetry

Read:

- Figure 1 shows the completed NTHS in 2007 over Chinese county boundaries and targeted city nodes.
- The expressway grid clearly links China's major metropolitan centers and, as a by-product, passes many smaller interior counties.
- Table 1 shows that connected peripheral counties were not random: compared with non-connected peripheral counties, they had higher GDP, higher GDP per capita, larger urban populations, and a lower agricultural output share already in 1997.

Economic message. The figure shows why the design is interesting, and the table shows why naive OLS is not enough.



FIGURE 1

China's National Trunk Highway System. The figure shows Chinese county boundaries in 1999 in combination with the targeted city nodes and the completed expressway routes of the NTHS in the year 2007.

auctions. Given the progress in the construction of the NTHS ahead of plan, the State Council approved an even more ambitious follow-up blue print for highway construction in 2004. The so called "7-9-18" system has the stated objective to connect all cities with an urban registered

(a) Figure 1

TABLE 1
Descriptive statistics for 1997

	Targeted city centres	Connected periphery	Non-connected periphery	National share of targeted city centres
Population (10,000)	233.24	56.96	38.48	0.17
Urban population (10,000)	179.69	10.77	5.83	0.5
GDP (100 Million Yuan)	517.86	32.58	15.09	0.5
GDP per capita (Yuan)	21435.06	5142.16	3637.09	—
Local government revenue (100 Million Yuan)	38.23	1.23	0.57	0.67
Industrial gross value added (100 Million Yuan)	194.61	14.93	5.58	0.48
Nonagricultural gross value added (100 Million Yuan)	505.75	24.42	9.74	0.59
Agricultural output share	0.04	0.34	0.42	—
Land area (km ²)	1543.09	3057.47	4513.4	0.015
Number of counties	54	424	943	54

Notes: The first three columns present mean 1997 levels, and the fourth column presents national shares by county groups. Targeted city centres refer to the central city county units (shixiaqu) of targeted metropolitan regions. Peripheral counties are counties outside a 50 km commuting buffer around the targeted city centres.

(b) Table 1

Figure 1: China's National Trunk Highway System and 1997 descriptive statistics.

5.2 Figures 2 and 3: how the instruments are built

Read:

- Figure 2 overlays the actual NTHS with the least-cost-path spanning tree. The least-cost network often tracks the broad geography of the actual network but is mechanically built from land cover and elevation costs.
- Figure 3 shows the Euclidean spanning tree, which is less geographically precise but captures a larger set of bilateral links actually chosen by planners.

Economic message. The paper's IV is not a black box. These figures make clear that the instruments are explicit counterfactual route maps based on cost minimization.

Both instruments correspond to the question of which routes central planners would have been likely to construct if the sole policy objective had been to connect all targeted destinations on a single network subject to global construction cost minimization. As I outline below, the least cost path network yields more precise route predictions between any given bilateral connection on an all-China minimum spanning tree due to its use of land cover and elevation data, while the Euclidean network covers a larger set of the actually built bilateral network routes.



FIGURE 2
Least cost path spanning tree network. The network in red colour depicts the completed NTHS network in 2007. The network in black colour depicts the least cost path spanning tree network. The black routes are the result of a

(a) Figure 2

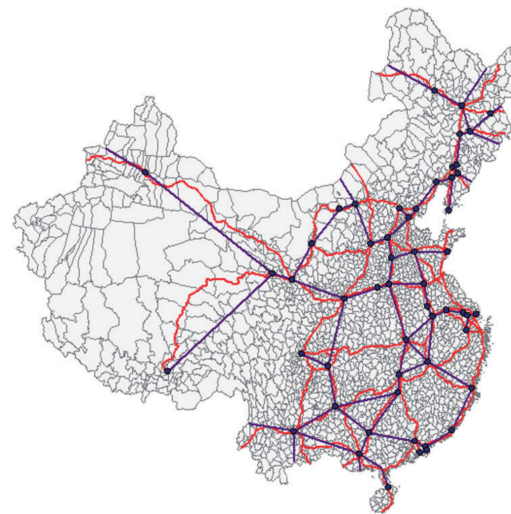


FIGURE 3
Euclidean spanning tree network. The network in red colour depicts the completed NTHS network in 2007. The network in darker colour depicts the Euclidean spanning tree network. The routes are the result of applying Kruskal's (1956) minimum spanning tree algorithm to bilateral Euclidean distances between targeted destinations. This algorithm is first run for the all-China network, and then repeated within North-Centre-South and East-Centre-West divisions of China. These regional repetitions add 9 routes to the original minimum spanning tree.

The following paragraphs describe the spanning tree instruments, and a more detailed description is provided in the Supplementary Appendix. The least cost path network depicted in [Figure 2] is constructed in a two-step procedure. The first step is to compute least cost highway construction paths between all possible targeted destination pairs on the basis of remote sensing data on land cover and elevation. To this end, I adopt a simple construction cost function from the transport engineering literature that assigns higher construction costs

(b) Figure 3

Figure 2: Least-cost-path and Euclidean spanning-tree instruments.

5.3 Table 2: first stage

Read:

- The least-cost-path and Euclidean instruments both significantly predict actual NTHS exposure.
- The first-stage F-statistics are strong in the main binary-treatment regressions.
- The instruments remain significant even when entered together, which suggests they capture related but not identical components of actual route placement.

Economic message. Table 2 shows relevance. Without it, the rest of the paper would not get off the ground.

TABLE 2
First stage regressions

Dependent variable:	(1) Connect	(2) Connect	(3) Connect	(4) lnDistHwy	(5) lnDistHwy	(6) lnDistHwy
Least cost path IV	0.323*** (0.0574)		0.254*** (0.0635)	0.317*** (0.0645)		0.245*** (0.0635)
Euclidean IV		0.243*** (0.0529)	0.144** (0.0560)		0.280*** (0.0599)	0.193*** (0.0657)
lnDistNode	-0.130*** (0.0376)	-0.127*** (0.0295)	-0.104*** (0.0323)	0.588*** (0.130)	0.635*** (0.112)	0.426*** (0.122)
Prefect capital	-0.124* (0.0648)	-0.129* (0.0736)	-0.120* (0.0658)	0.437* (0.209)	0.429* (0.229)	0.413* (0.215)
City Status	0.0891** (0.0403)	0.0929** (0.0437)	0.0847** (0.0399)	-0.297*** (0.0946)	-0.296*** (0.103)	-0.270*** (0.0951)
lnUrbPop90	0.106*** (0.0225)	0.115*** (0.0217)	0.107*** (0.0209)	-0.228*** (0.0691)	-0.244*** (0.0640)	-0.227*** (0.0636)
Educ90	-0.273 (0.598)	-0.303 (0.656)	-0.302 (0.601)	-1.671 (1.697)	-1.747 (1.804)	-1.626 (1.666)
AgShare90	-0.170 (0.182)	-0.216 (0.189)	-0.167 (0.179)	0.0238 (0.537)	-0.00173 (0.555)	-0.0160 (0.533)
Obs	1342	1342	1342	1342	1342	1342
R ²	0.222	0.204	0.233	0.401	0.394	0.414
First stage F-Stat	31.61	21.07	20.31	24.09	21.82	15

Notes: All regressions include province fixed effects. Columns 1–3 report results for binary NTHS connection indicators among peripheral counties. Columns 4–6 report results for the log distance to the nearest NTHS segment among peripheral counties. lnDistNode is log county distance to the nearest targeted city node. Prefect Capital and City Status are binary indicators for respective county status in 1990. lnUrbPop90 is log 1990 county urban population. Educ90 is the 1990 county share of above compulsory schooling in 20+ population. AgShare90 is the 1990 county share of agricultural employment. Standard errors are clustered at the province level and stated in parentheses below point estimates. ***1%, **5%, and *10% significance levels.

Figure 3: Table 2 from the paper: first-stage regressions.

5.4 Table 3: main reduced-form and IV results

Read:

- OLS coefficients are negative but often small and insignificant.
- IV coefficients are materially more negative, especially after adding the full set of county controls.
- In the preferred specification, connected peripheral counties have significantly lower growth in industrial output, nonagricultural output, local government revenue, and GDP.
- Agricultural output and population are not significantly affected.

Economic message. This is the core result: connecting the periphery to the core lowered relative industrial and GDP growth in the periphery.

TABLE 3
Network connection effects among peripheral counties

Variables		(1) OLS No controls	(2) OLS With controls	(3) LCP IV No controls	(4) LCP IV With controls	(5) Euclid IV No controls	(6) Euclid IV With controls	(7) Both IVs No controls	(8) Both IVs With controls	(9) Both IVs With controls
GVA)	Connect	-0.0529 (0.0418)	-0.0356 (0.0499)	-0.284** (0.118)	-0.304** (0.145)	-0.246* (0.148)	-0.287* (0.154)	-0.272*** (0.0965)	-0.297*** (0.108)	-0.297** (0.121)
	Obs	1302	1280	1302	1280	1302	1280	1302	1280	1280
	R ²	0.242	0.255							
Δ)	Connect	-0.0411 (0.0335)	-0.0266 (0.0375)	-0.243** (0.0983)	-0.252** (0.117)	-0.270** (0.122)	-0.296** (0.131)	-0.251*** (0.0877)	-0.268*** (0.0969)	-0.268*** (0.0946)
	Obs	1285	1262	1285	1262	1285	1262	1285	1262	1262
	R ²	0.27	0.284							
te)	Connect	-0.0497* (0.0285)	-0.0914*** (0.0295)	-0.0542 (0.109)	-0.223* (0.120)	-0.175 (0.117)	-0.315** (0.132)	-0.0926 (0.0893)	-0.257*** (0.0996)	-0.257*** (0.100)
	Obs	1290	1285	1290	1285	1290	1285	1290	1285	1285
	R ²	0.275	0.334							
P)	Connect	-0.00204 (0.0245)	-0.0144 (0.0276)	-0.106 (0.0830)	-0.177* (0.0942)	-0.178 (0.112)	-0.254** (0.116)	-0.127 (0.0824)	-0.203** (0.0886)	-0.203** (0.080)
	Obs	1297	1272	1297	1272	1297	1272	1297	1272	1272
	R ²	0.228	0.264							
GVA)	Connect	-0.00344 (0.0210)	-0.00790 (0.0220)	0.000194 (0.0631)	-0.0252 (0.0789)	-0.0305 (0.0672)	-0.0597 (0.0728)	-0.00865 (0.0545)	-0.0371 (0.0630)	-0.0371 (0.0654)
	Obs	1335	1313	1335	1313	1335	1313	1335	1313	1313
	R ²	0.202	0.208							
ulation)	Connect	0.00488 (0.00456)	-0.00217 (0.00568)	0.0395** (0.0188)	0.0264 (0.0234)	0.0183 (0.0242)	0.0104 (0.0262)	0.0333* (0.0183)	0.0207 (0.0215)	0.0207 (0.0225)
	Obs	1337	1314	1337	1314	1337	1314	1337	1314	1314
	R ²	0.234	0.271							

oint estimate stems from a separate regression. All regressions include province fixed effects. LCP IV stands for the least cost path spanning tree instrument. Euclid IV straight line spanning tree instrument. "No controls" columns refer to regressions on NTHS treatment and log county distance to the nearest targeted city node. With atee a full set of 1000 county controls (city status and prefecture capital dummies, log urban population, share of agricultural employment, and share of above compulsory

Figure 4: Table 3 from the paper: network connection effects among peripheral counties.

5.5 Table 4: placebo before and after the network was built

Read:

- Before the network was built, the connection measure does not predict local-government-revenue growth.
- After the network is built, the same connection measure becomes negative and significant.
- The same pattern appears in the distance specification: no effect pre-network, positive and significant distance effect post-network.

Economic message. The instrument is not just finding counties that were already on a worse trajectory before the road shock.

TABLE 4
Falsification test before and after the network was built

Variable:	(1) OLS 1990–97	(2) OLS 1997–06	(3) LCP IV 1990–97	(4) LCP IV 1997–06	(5) Euclid IV 1990–97	(6) Euclid IV 1997–06	(7) Both IVs 1990–97	(8) Both IVs 1997–06
County	0.0154 (0.0410) 894 0.274	−0.0848** (0.0360) 894 0.339	0.0143 (0.0853) 894	−0.151 (0.0974) 894	0.117 (0.107) 894	−0.282** (0.129) 894	0.0563 (0.0647) 894	−0.204*** (0.0467) 894
Distance	−0.0114 (0.0142) 894 0.275	0.0160 (0.0190) 894 0.336	−0.0409 (0.0350) 894	0.0854* (0.0470) 894	−0.00442 (0.0573) 894	0.185** (0.0783) 894	−0.0274 (0.0329) 894	0.122*** (0.0430) 894
Constant			19.635	19.635	19.091	19.091	14.93	14.93
Distance								
Constant			18.696	18.696	17.306	17.306	11.259	11.259

Joint estimate stems from a separate regression. All regressions include province fixed effects and a full set of county controls. LCP IV stands for the least cost path instrument. Euclid IV stands for the straight line spanning tree instrument. Panel A presents results for binary NTHS connection indicators (for both OLS and instruments) and presents results for log distance to the nearest NTHS segment (again for both OLS and instruments). Standard errors are clustered at the province level and stated in parentheses. ***1%, **5%, and *10% significance levels.

Figure 5: Table 4 from the paper: falsification test before and after the network was built.

5.6 Table 5 and Figure 4: channels and spatial decay

Read:

- Table 5 shows no significant effect on urban population or urbanization.
- The negative connection effect on industrial output, GDP, and local-government revenue remains when the paper adds a dummy for neighboring peripheral counties.
- The neighbor effect itself is small and insignificant.
- Figure 4 shows that the adverse effect fades smoothly with distance from the highway, with no non-monotonic pattern around neighboring counties.

Economic message. These results weigh against a decentralization story and in favor of a trade-based mechanism.

TABLE 5
Are NTHS connections associated to urbanization and industrial decentralization among peripheral counties?

Variable:	Change ln(UrbPop) 1997–06 (1)	Change ln(Urb/Pop) 1997–06 (2)	Change ln(IndGVA) 1997–06 (3) (4)		Change ln(GDP) 1997–06 (5) (6)		Change ln(GovRevenue) 1997–06 (7) (8)		Change ln(GovRevenue) 1997–06 (9) (10)		(11)
	0.0350 (0.0953)	0.0137 (0.0925)	–0.297*** (0.108)	–0.262** (0.113)	–0.203** (0.0886)	–0.193** (0.0919)	–0.257*** (0.0996)	–0.253*** (0.0961)	0.0535 (0.195)	0.177*** (0.0667)	
				0.153 (0.214)	0.113* (0.0615)		0.0907 (0.132)	0.0845* (0.0480)			
Stat	13.374 1,072	13.374 1,072	18.886 1,280	5.016 1,280	13.852 1,280	17.425 1,272	4.84 1,272	12.989 1,272	19.055 1,285	5.383 1,285	13.879 1,285

Regressions include province fixed effects and a full set of county controls. Reported results are 2nd stage estimates using the least cost path and the Euclidean spanning tree instrument for NTHS connections, neighbouring peripheral counties, or distance to the nearest NTHS segment. Columns 1 and 2 report connection effects on peripheral counties in log urban population and urbanization, respectively. Neighbour indicates peripheral counties neighbouring a connected peripheral county. Standard errors are clustered at the county level and stated in parentheses below point estimates. ***1%, **5%, and *10% significance levels.

Figure 6: Table 5 from the paper: urbanization and industrial-decentralization tests.

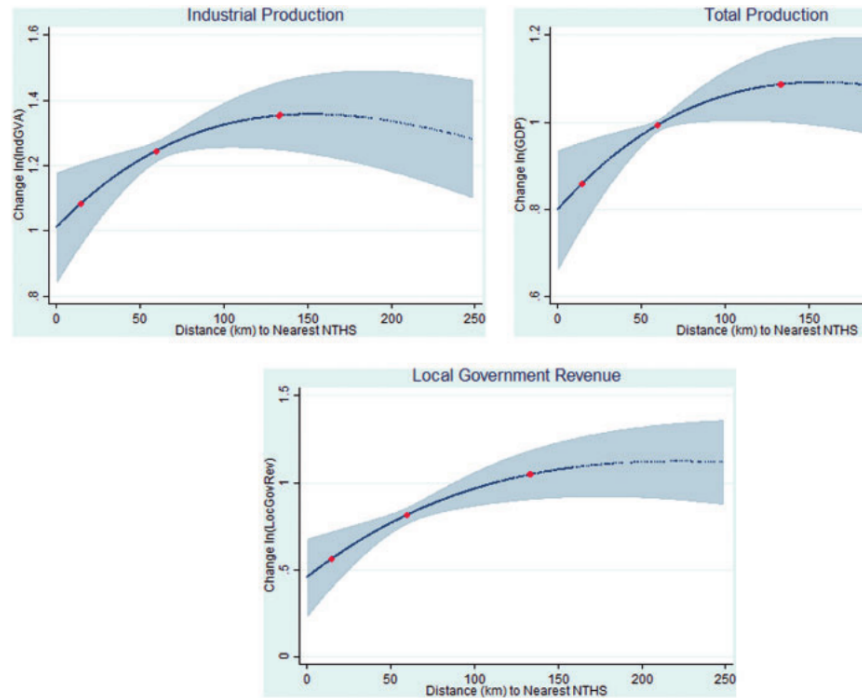


FIGURE 4

Estimated effect of peripheral connections over distance to the nearest NTHS route. The graphs depict estimated relationships between distance to the nearest NTHS route and peripheral county growth in industrial, total GDP, and local government revenue. The plots correspond to the best fitting polynomial function according to the Akaike Information Criterion (AIC). The functions and confidence intervals are based on holding covariates at their mean. County distance to the NTHS and its polynomial terms are instrumented using distances to the LCP and Euclidean spanning trees and their polynomials. The red dots indicate median county distance to the nearest NTHS route among connected peripheral counties (left), peripheral counties neighbouring county (centre), and the remaining peripheral counties farther away (right). The shaded areas indicate 90% confidence intervals. Standard errors are clustered at the province level.

Figure 7: Figure 4 from the paper: adverse effects decline with distance to the NTHS.

5.7 Table 6: heterogeneity by initial size and remoteness

Read:

- The interaction of the connection dummy with $\ln DistNode$ is positive and significant.
- The interaction of the connection dummy with the above-median-employment indicator is also positive and significant.
- In the continuous-distance version, the interaction terms are negative and significant.

Economic message. Larger and more remote peripheral counties are less badly hit by the fall in trade costs, exactly as a core-periphery trade model predicts.

TABLE 6
Testing the heterogeneity of peripheral connection effects

Dependent variable:	Change ln(IndGVA) 1997–2006		Change ln(GDP) 1997–2006	
	(1)	(2)	(1)	(2)
<i>Panel A: Binary</i>				
Connect	−0.304** (0.145)	−4.281*** (1.569)	−0.177* (0.0942)	−3.571*** (1.011)
Connect*ln(DistNode)		0.748*** (0.270)		0.636*** (0.172)
Connect*Emp90Dum		0.450* (0.255)		0.404** (0.196)
Obs	1280	1280	1272	1272
First stage F-Stat	29.966	3.462	27.972	4.724
<i>Panel B: log Distance</i>				
lnDistHwy	0.0954 (0.0674)	1.465*** (0.455)	0.0639 (0.0434)	1.105*** (0.318)
lnDistHwy*ln(DistNode)		−0.236*** (0.0748)		−0.181*** (0.0494)
lnDistHwy*Emp90Dum		−0.266*** (0.0823)		−0.192*** (0.0693)
Obs	1280	1280	1272	1272
First stage F-Stat	22.367	4.649	21.698	4.842

Notes: All regressions include province fixed effects and a full set of county controls. Reported results are 2nd stage estimates using the LCP spanning tree to instrument for NTHS connections as well as their reported interaction terms. lnDistNode is log county distance to the nearest targeted city node. Emp90Dum is a dummy for counties with above mean levels of county employment in 1990. Standard errors are clustered at the province level and stated in parentheses below point estimates. ***1%, **5%, and *10% significance levels.

The reported results provide empirical evidence of significant heterogeneity in the NTHS

Figure 8: Table 6 from the paper: heterogeneity of peripheral connection effects.

6 Short summary

- The paper studies whether a massive transport network helped China's periphery once it was connected to the metropolitan core.
- It uses county-level data and an IV strategy based on minimum spanning tree networks to isolate plausibly exogenous variation in exposure to the NTHS.
- The baseline finding is stark: connected peripheral counties experienced lower industrial, non-agricultural, revenue, and GDP growth than comparable non-connected peripheral counties.
- The estimates become more negative after adding county controls, which is consistent with planners having positively selected richer and more urbanized counties onto routes.
- Placebo tests show no corresponding pre-network effects.
- Mechanism tests reject a simple urbanization or local decentralization story and instead support a trade-based channel in which falling trade costs with very large core markets hurt small peripheral producers.

- Heterogeneity results fit the comparative statics of core-periphery theory: the negative effect is weaker for initially larger and more remote peripheral counties.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that large-scale inter-regional transport infrastructure can reduce, rather than increase, industrial and total output growth in connected peripheral regions when those regions are linked more closely to much larger metropolitan centers.

7.2 Economic intuition

Lower trade costs are not always an equalizing force. When market sizes are highly asymmetric, the core may gain relatively more from integration because it has stronger agglomeration forces, better market access, and a larger demand base. The periphery may gain access to the core, but it also becomes more exposed to core competition.

7.3 Takeaway

Faber's main lesson is that infrastructure policy should not be evaluated only through national efficiency gains or the engineering logic of connectivity. The general-equilibrium spatial consequences matter. Roads that connect the periphery to the core may help the country as a whole while still weakening industrialization in the periphery itself.